XRPL Validator — Daily Engineering Report · 2026-07-31

Period: 2026-07-31 00:00 to 24:00 US Eastern (2026-07-31 04:00 to 2026-08-01 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-31): 0.78788 (2,968 missed of 13,992). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ledger): CredentialDelete needs only one of Subject/Issuer** (`aaf0c79`, xrpl-validator) Our preflight required BOTH Subject and Issuer. rippled requires at least ONE (CredentialDelete.cpp preflight: `if (!subject && !issuer) return temMALFORMED`) and resolves the pair in preclaim with
- **fix(ledger): funding outranks expiry in OfferCreate preclaim** (`5582a93`, xrpl-validator) rippled's OfferCreate::preclaim (:171) runs
- **fix(ledger): enforce lsfRequireAuth on the receiving trust line** (`a67eeca`, xrpl-validator) We never modelled lsfRequireAuth in the liquidity path at all — only in trust_set.rs for line-deletion decisions. rippled enforces it at STRAND CONSTRUCTION (DirectStep.cpp:430-437, DirectIPaymentStep::check): auto const authField = (src_ > dst_) ? lsfHighAuth : lsfLowAuth; if (sleSrc->isFlag(lsfRequireAuth) && !sleLine->isFlag(authField) && (*sleLine)[sfBalance] == beast::kZero) return terNO_AUTH;
- **fix(ledger): remove the taker's own offer even when nothing crosses** (`70ea97e`, xrpl-validator) rippled deletes the taker's OWN offer off the tip of the book during offer crossing, and its commentary spells out why (BookStep.cpp:415-455): "We could skip over the self offer in the book and only cross offers that are not our own. This would make a lot of sense, but we don't do it. Part of the rationale is that we can only operate on the tip of the order book. We can't leave an offer behind -- it would sit on the …
- **fix(ledger): tfOneAssetWithdrawAll empties the LP position** (`e12436b`, xrpl-validator) rippled's isWithdrawAll covers BOTH tfWithdrawAll (0x00020000) AND tfOneAssetWithdrawAll (0x00040000) — AMMWithdraw.cpp:1133-1138: if ((tx[sfFlags] & (tfWithdrawAll | tfOneAssetWithdrawAll)) != 0u) return WithdrawAll::Yes;

Docs and operations

- **Revert "fix(ledger): remove the taker's own offer even when nothing crosses"** (`a0363a7`, xrpl-validator)

Independence program

No independence record for UTC day 2026-07-31; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Stealth Meta-Address Registry for Payments** (privacy) Add a new owner-reserved ledger object `StealthKey` (keylet::stealthKey(account)) holding sfScanPublicKey and sfSpendPublicKey (33-byte secp256k1 poin
- **Oracle-Linked Dynamic NFT Valuation Sync** (nft) Extend XLS-46's mutable-NFT mechanism with a new mint-time flag tfOracleLinked (valid only alongside tfMutable) and three immutable SFields on the NFT
- **MPT Flow-Rate Payment Channels** (payments) Extend PayChan to accept an MPT-denominated Amount (sfMPTokenIssuanceID alongside the existing XRP path) at PaymentChannelCreate, and add two SFields

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-31 15:51	—	xrpl-validator	`aaf0c79`	fix(ledger): CredentialDelete needs only one of Subject/Issuer
07-31 15:51	—	xrpl-validator	`5582a93`	fix(ledger): funding outranks expiry in OfferCreate preclaim
07-31 16:00	—	xrpl-validator	`a67eeca`	fix(ledger): enforce lsfRequireAuth on the receiving trust line
07-31 17:20	—	xrpl-validator	`70ea97e`	fix(ledger): remove the taker's own offer even when nothing crosses
07-31 17:42	—	xrpl-validator	`a0363a7`	Revert "fix(ledger): remove the taker's own offer even when nothing crosses"
07-31 19:20	—	xrpl-validator	`e12436b`	fix(ledger): tfOneAssetWithdrawAll empties the LP position

Times in UTC. A commit appears on the day it was committed to main or merged into it.